

Combinatorial chemistry yields a learnable transcriptomic design map in Z-Screen

Zafrens, Inc.

Abstract

A phenotypic screen produces substantially more value when its output functions as a design map (a function from chemistry to reproducible biological program that a discovery team can interrogate, reuse, and extend) rather than as a static catalog of measurements. The Z-Screen platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per microwell. Every well therefore carries full chemistry provenance and a structured transcriptomic readout. Across 615,793 repaired RNA profiles spanning 12 combinatorial libraries and 4 cell lines (615,721 with valid scVI latent coordinates), the canonical dataset already supports the design-map transition at pilot scale and leaves substantial headroom for expansion.

The assay floor is high across the main systems. Median split-half cosine for control compounds is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. On held-out tuples within a library, building-block models substantially outperform centroid baselines. The strongest case is ZEL024 in HEK293, where a building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline (33.2% reduction in mean squared error). ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). Chemistry-only descriptors carry predictive signal as well: in ZEL024 / HEK293, a combined building-block plus chemistry-embedding model reaches cosine 0.621 with a 29.3% error reduction, the highest gain among any chemistry-only configuration evaluated.

The dataset also produces direct discovery-style signals. Background-calibrated mimicry against per-cell-line cross-control similarity gives the cleanest hit in ZEL024 / HEK293, where a library tuple matches the internal reference ZF-104 at cosine 0.901 against a HEK293 cross-control background of median 0.200 and 99th percentile 0.931. ZEL031 / THP1 yields tuples reaching cosine 0.942 to sorafenib, 0.926 to MZ1, and 0.918 to STC-15. THP1 has a higher cross-control background (median 0.589, 99th percentile 0.972), so those matches sit in the upper tail of plausible transcriptomic neighborhoods rather than uniquely on-mechanism territory. Building-block effects also recur across cell lines at chemistry-resolved positions (ZEL024 bb3 median 0.602 between HEK293 and H1650; ZEL031 bb0 median 0.355 and bb1 median 0.526 between A549 and THP1), with the strongest building blocks reaching cosine 0.87. Imaging adds an additional but smaller benefit in a simple additive benchmark, with image plus RNA reaching 0.540 balanced accuracy in ZEL024 controls

against 0.524 for RNA alone and 0.213 for image alone.

In aggregate, these results show that Z-Screen behaves as a chemistry-to-transcriptome modeling platform on data from a small number of pilot experiments. The core asset is the RNA branch and the chemical provenance behind it; imaging is directionally helpful and is expected to compound as image consistency improves; the platform is engineered to scale by orders of magnitude beyond the present dataset, so the next investment compounds a signal already present rather than testing whether one exists.

Significance

Most screening campaigns terminate as a ranked list of wells. The present analysis demonstrates that Z-Screen supports a different output: a design map. Because every Z-Screen compound carries its building-block history and its measured cell-state response, the platform can learn which structural elements move biology in useful directions, which combinations are redundant, and which compounds mimic known reference states. The relevant outcome is not that one model exceeds one benchmark; it is that chemistry, transcriptomics, and known-control behavior align well enough to support prediction, triage, and follow-up design. For a discovery organization, this converts a screen from a one-time search into reusable institutional memory.

Introduction

Early-stage discovery requires throughput and biological resolution simultaneously, and existing platforms force a compromise. High-content imaging is scalable and information-rich but mechanistically thin (Cell Painting, Recursion’s RxRx atlases, Insitro’s morphology platforms, the Broad’s JUMP consortium). Transcriptomic profiling provides closer access to underlying cell state at higher cost per compound (Perturb-seq [3], Tahoe-100M, scPerturb). Combining the two at scale on the same compounds has been difficult. Public efforts that have attempted this combination have reached treatment-matched scale rather than same-well: scGeneScope, the strongest current public benchmark, pairs imaging and RNA at the level of replicate populations exposed to the same compound, not the same biology [4].

The state-of-the-art chemistry-side platform is the Connectivity Map and LINCS L1000 program: approximately a million compound profiles in a deliberately reduced 1,000-gene transcriptional space, with bulk rather than single-cell readout [2]. L1000 set the standard for chemistry-to-transcriptome similarity search and remains the deepest public chemistry catalog by compound count. What L1000 does not provide is design information. A typical L1000 experiment is a single profiled compound with no internal structure to its identity. There is no built-in way to ask which specific structural element moved which biological program, because the dataset was built to compare compounds rather than to compose them.

Z-Screen was built around a different premise. A combinatorial chemistry screen produces substantially more value when every compound contributes a reusable biological program rather than a sparse binary label, and when each compound’s identity is decomposable into building-block

coordinates with full SMILES. The output then becomes a map from chemical composition to transcriptomic phenotype rather than a list of active compounds. With a stable map of that shape, a discovery team can rank unmeasured chemistry, identify building blocks that carry the strongest programs, flag compounds that mimic reference states, and reason about design rather than only about screening.

The analyses below characterize what the canonical dataset, itself assembled from a small number of pilot experiments, currently supports. Six concrete questions are addressed:

1. Are control compounds reproducible enough to support modeling?
2. Does building-block composition predict held-out transcriptomic states?
3. Does full chemical structure add predictive signal beyond building-block identity?
4. Do library tuples recover recognizable reference programs once cross-control similarity backgrounds are accounted for?
5. Do building-block effects recur across cell lines?
6. Does imaging contribute value in a restrained multimodal benchmark?

Each question is answerable on the present pilot. The aim is to establish what the current data already does well enough to justify a substantially larger campaign, with appropriate qualification of how much of the platform’s full capacity has been used so far.

Results

Control profiles establish a stable assay floor across cell lines

The chemistry-prediction question is only meaningful if the assay itself is stable enough to support it. Split-half reproducibility for named controls is high across the main systems. Median control cosine is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. THP1 sits below the others because its compound-control library coverage is sparser, not because the readout is weaker on monocytes.

These numbers leave clear headroom between technical reproducibility and current model performance, which is relevant when interpreting the chemistry models below: a model reaching cosine 0.674 retains room for improvement because the readout itself is highly consistent.

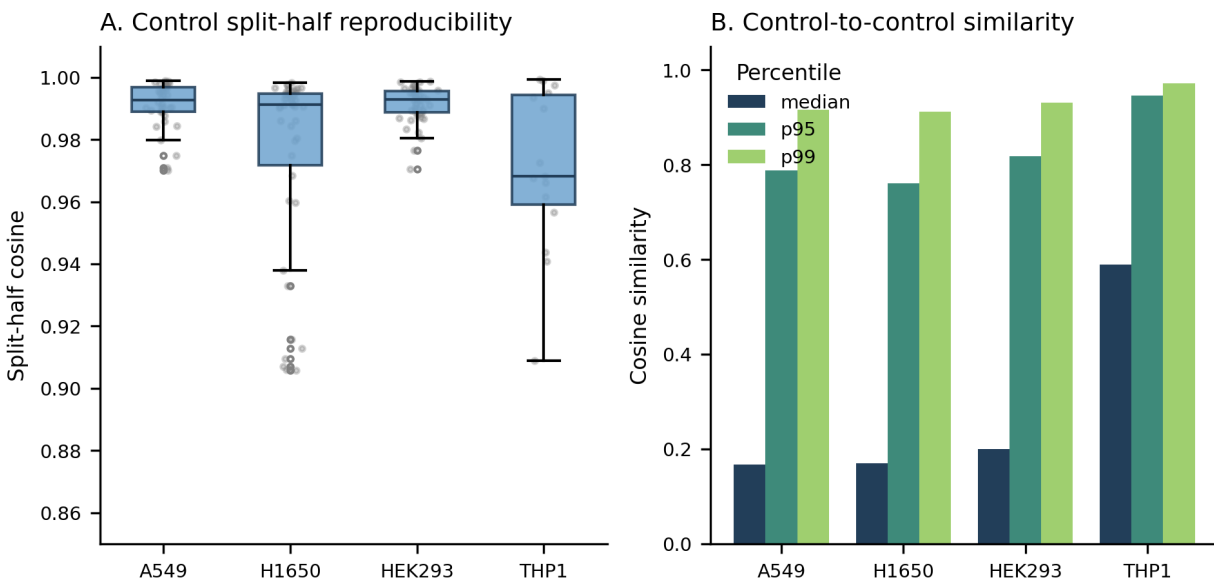


Figure 1. Named controls show high split-half reproducibility across all four cell lines, establishing a strong assay floor for downstream chemistry-to-phenotype modeling.

Building-block models predict held-out compound programs

The central modeling question is whether compositional chemistry carries enough signal to predict the transcriptomic state of a held-out library member. Across several library and cell-line systems, the answer is yes.

ZEL024 in HEK293 is the strongest case. A building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline, a 33.2% reduction in mean squared error. ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). ZEL031 in A549 and ZEL024 in H1650 also showed positive gains, although less pronounced.

The model does more than memorize whole compounds. It learns that particular chemical pieces and combinations of pieces carry recurring transcriptomic programs, which supports design-time reasoning about combinations not yet screened. Within the broader generalization-ladder framework laid out in companion paper³, this benchmark is interpolation within the shared building-block vocabulary of each library; the harder building-block holdout and scaffold-family transfer regimes are evaluated separately there.

Predicting unseen transcriptomes from chemistry

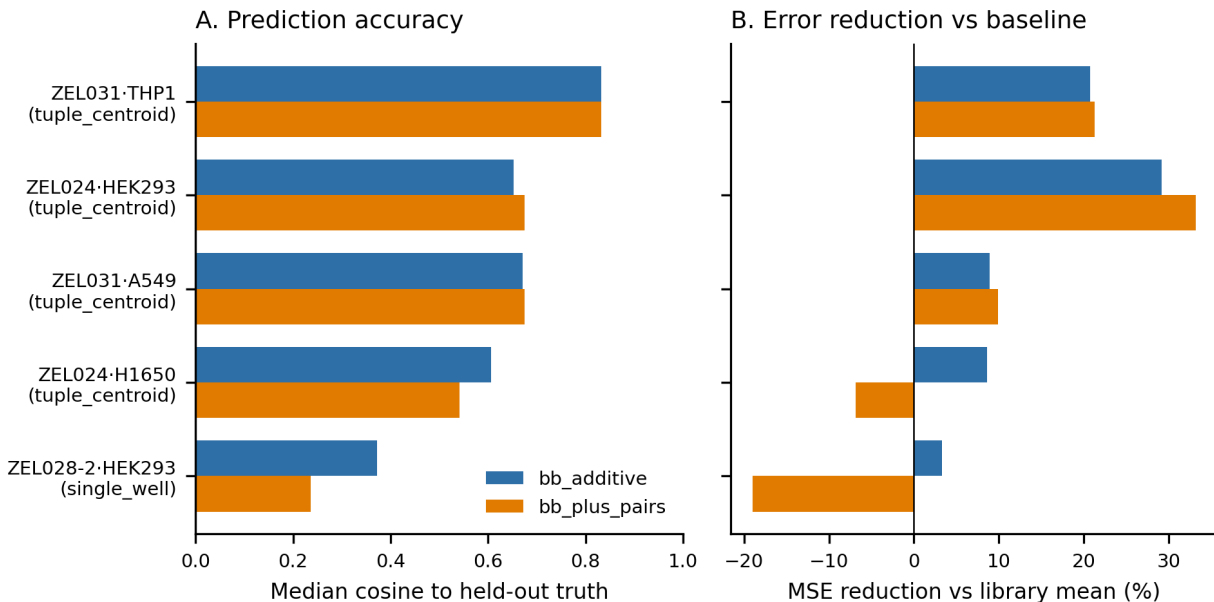


Figure 2. Building-block models improve prediction of held-out transcriptomic states across multiple library and cell-line systems, with the strongest gains in ZEL024 / HEK293 and ZEL031 / THP1.

Full chemical structure adds design-time predictive signal

Building-block identity is the strongest single signal in this dataset, and full structure adds further signal in the strongest ZEL024 system. In ZEL024 / HEK293, chemistry-only models were already predictive: a 256-dim chemistry embedding alone (a private random projection of ECFP4 fingerprints, see Methods) reached cosine 0.611 (27.8% error reduction), building-block identity alone reached 0.611 (27.5% error reduction), and combining the two reached 0.621 (29.3% error reduction).

This pattern is consistent with a future design loop. Building-block provenance reflects the combinatorial grammar of the library and remains valuable. Full structure carries additional information beyond those labels, so adding it improves prediction further. The combined representation is therefore the natural input for ranking unmeasured compounds by expected transcriptomic effect. Companion paper³ quantifies how much of the chemistry-only gain reflects interpolation within a library and how much survives stricter held-building-block evaluation.

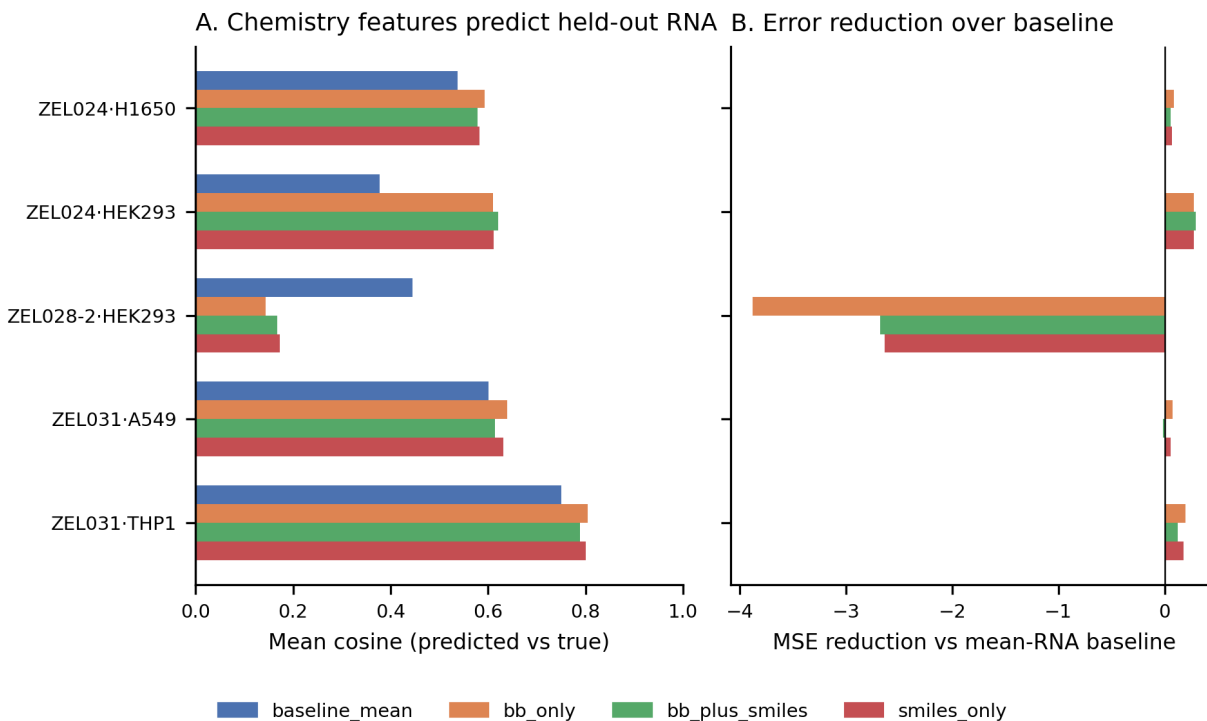


Figure 3. Chemistry-only models capture predictive signal, and the combination of building-block provenance with full-structure descriptors performs best in the strongest ZEL024 system.

Library tuples recover recognizable reference pharmacology states

Control mimicry is the most direct discovery result the dataset produces: library tuples whose transcriptomic centroids land near the centroid of a known reference compound. The cleanest mimicry result by background-calibrated standards comes from ZEL024 / HEK293, where a library tuple matched the internal reference ZF-104 at cosine 0.901. The HEK293 between-control similarity background sits at median 0.200 with a 99th percentile of 0.931, which places a 0.901 match for an unrelated chemistry tuple in the upper tail of plausible transcriptomic neighborhoods rather than the bulk. Several additional ZEL024 / HEK293 tuples reached cosine 0.85 to 0.89 against published references such as ZEL029-25 and ZEL029-28.

Mimicry hits also appear in ZEL031 / THP1, with tuples reaching cosine 0.942 to sorafenib, 0.926 to MZ1, and 0.918 to STC-15. THP1 is useful for several pharmacology programs, but its cross-control similarity background is substantially higher (median 0.589, 99th percentile 0.972). The THP1 hits are biologically meaningful and populate the upper tail of plausible neighborhoods, but they should not be promoted as uniquely on-mechanism without orthogonal validation. The THP1 background is reported explicitly so the calibration is visible to readers.

The screen produces new chemistry that lands in recognizable biological neighborhoods, with the strongest defensible discovery result coming from a cell line where the cross-control background is low enough to support the mimicry claim without further qualification.

Library tuples vs nearest control compound

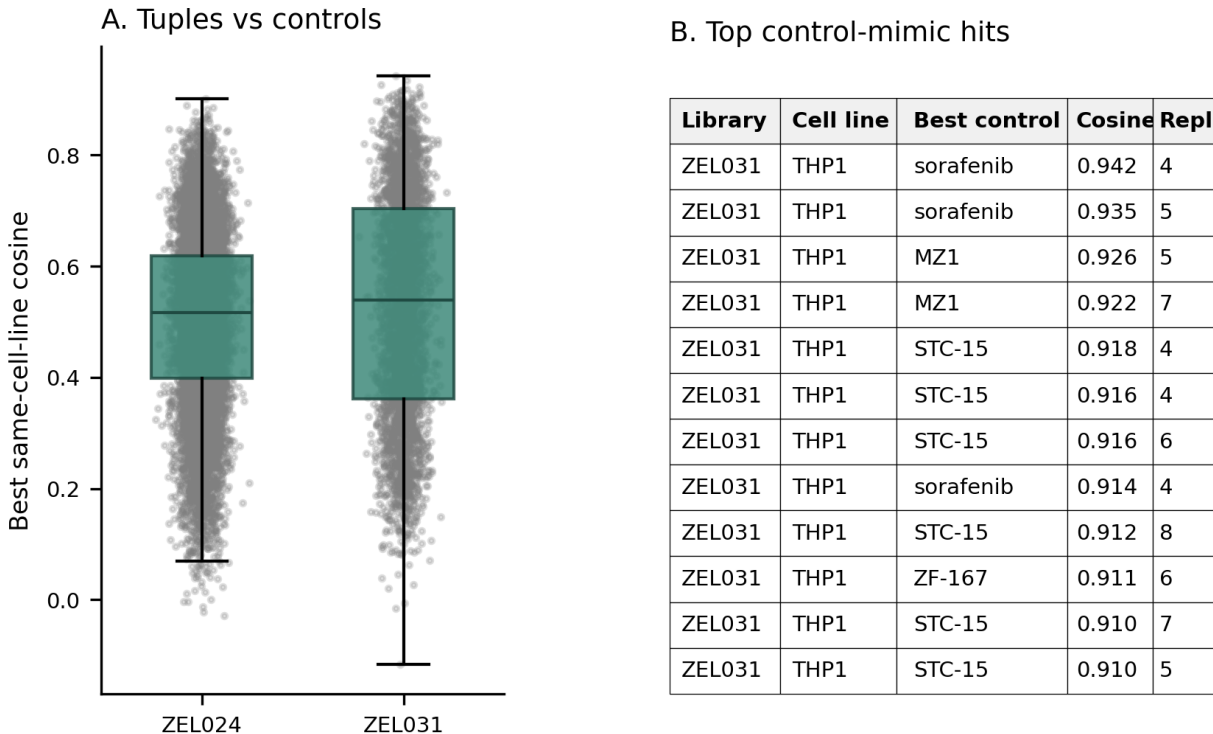


Figure 4. Multiple library tuples land near transcriptomic neighborhoods centered on known reference compounds, with the strongest background-calibrated match in HEK293 (ZF-104 at cosine 0.901, against a HEK293 99th-percentile cross-control background of 0.931) and additional THP1 hits to sorafenib, MZ1, and STC-15 in a system with a higher cross-control baseline.

Building-block programs recur across cell lines

A reusable chemistry model is more valuable when its building-block programs survive a change in cellular context. Per-building-block transcriptomic programs were compared in matched library systems across paired cell lines. Results are reported by building-block position because positions with very few unique values (e.g. ZEL024 bb0, bb1, bb2) collapse hundreds-to-thousands of distinct chemistries into a single average and therefore measure library-wide rather than chemistry-resolved transfer. The chemistry-resolved positions are ZEL024 bb3 (83 building blocks, median 168 unique tuples averaged per signature) and ZEL031 bb0 / bb1 (a 2-position library, where each single-position signature still leaves the other position to vary across approximately 70 to 100 distinct partner chemistries).

Across these chemistry-resolved positions, building-block programs partially survive cell-context change. In ZEL024, the median cross-cell cosine at bb3 between HEK293 and H1650 was 0.602 ($n = 83$, IQR 0.51 to 0.69, strongest building blocks reaching 0.873). In ZEL031 between A549 and THP1, the median cross-cell cosine was 0.355 at bb0 ($n = 105$, IQR 0.23 to 0.48, strongest 0.747)

and 0.526 at bb1 ($n = 84$, IQR 0.32 to 0.67, strongest 0.808).

These medians sit meaningfully below the within-cell reproducibility ceiling, so the result is not a claim of perfect transfer. The defensible interpretation is that many building blocks carry partially conserved biological directionality across cell types, which is the prerequisite for reusing chemistry across programs. The ZEL024 bb0 / bb1 / bb2 cross-cell results are not chemistry-resolved at this library architecture (each averaged signature reflects 900 to 7,000 distinct tuples) and are therefore reported as a diagnostic in `paper2/tables/bb_effect_consistency.csv` rather than as a headline claim. Companion paper4 extends the cross-cell question to the molecule and grouped-program level, where the transfer story strengthens substantially in chemistry-resolved settings.

Shared BB program consistency across cell lines

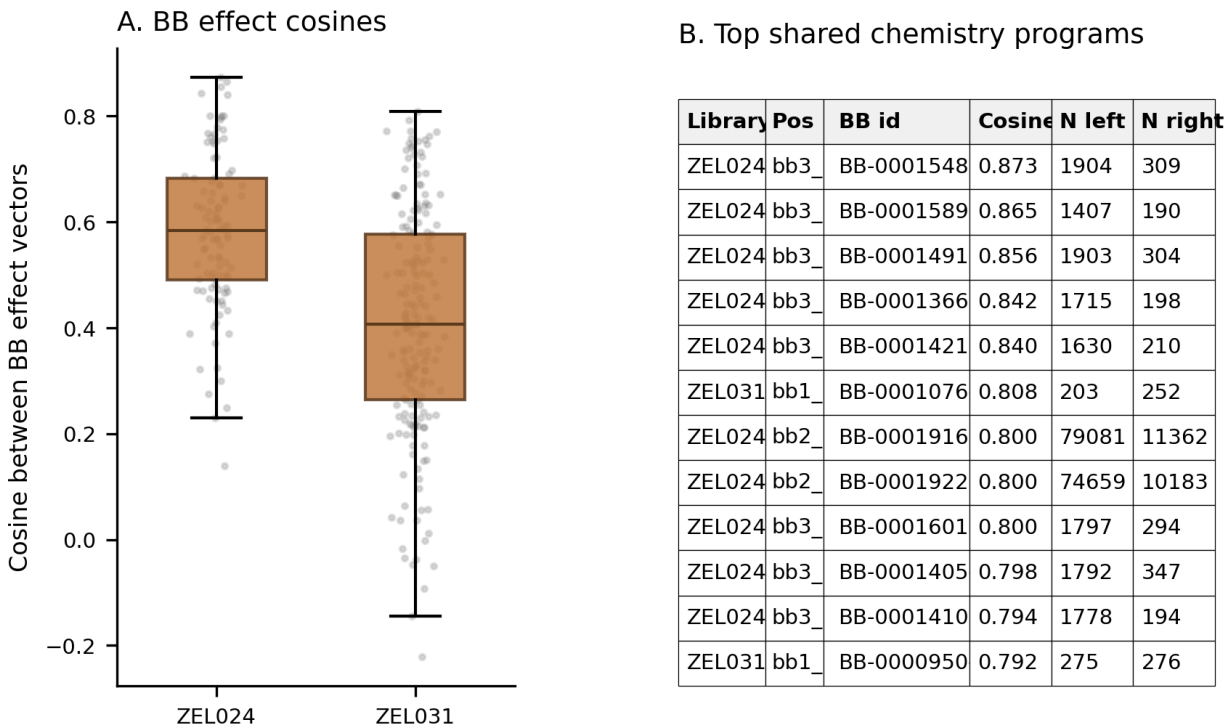


Figure 5. Building-block effects recur across cell lines, with clear positive median transfer and a subset of building blocks showing strong cross-context conservation.

Imaging adds signal in a simple multimodal benchmark

The current platform story is RNA-first, and the image branch already adds value when framed conservatively. In the strongest additive multimodal benchmark, based on ZEL024 controls, image-only classification reached 0.213 balanced accuracy, RNA-only reached 0.524, and the combined representation reached 0.540. The smaller ZEL031 recurrent-tuple benchmark followed the same ordering at lower absolute values: 0.127 for image, 0.146 for RNA, and 0.180 for the combined model. Companion paper1 shows a substantially larger image contribution in the same-well ActiveSeq

matched-readout setting after task-aware calibration of the per-well image latent, indicating that the modest gain reported here reflects current image-pipeline maturity rather than a ceiling on what same-well multimodality can achieve in Z-Screen.

The multimodal claim at the present pilot is measured. Imaging is not the lead modality today, the combined representation is consistently better than either single view in a practical classification task, and continued investment in the imaging workflow should compound that gain.

ActiveSeq-style additive multimodal probe

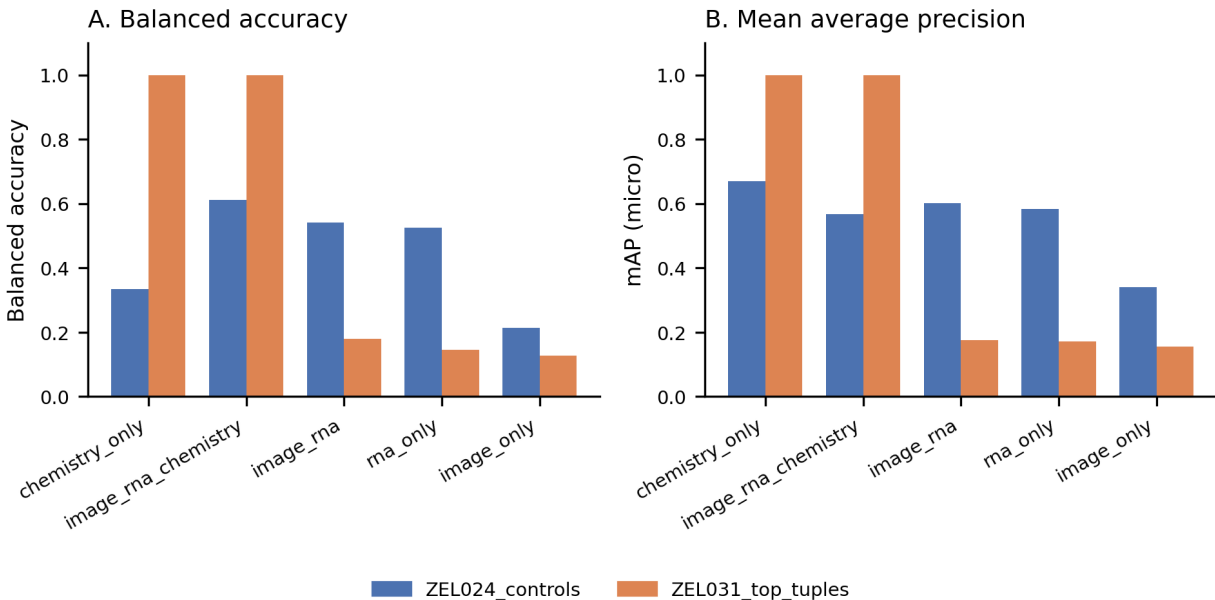


Figure 6. In a simple additive benchmark, image plus RNA outperforms either single modality alone, while RNA remains the dominant branch in the current dataset.

External benchmark context: position relative to Tahoe-100M, scGeneScope, and L1000

For external positioning, the present pilot is most usefully compared with four public reference points. Tahoe-100M is the scale anchor for perturbational transcriptomics, with over 95.6 million RNA profiles across 50 cancer cell lines and approximately a thousand small-molecule perturbations [3]; it does not include matched imaging and was not built around a combinatorial chemistry framework. scGeneScope is the closest public multimodal benchmark, with 627,704 RNA profiles and 716,767 images across 28 chemical treatments in U2-OS cells, paired at the level of replicate populations rather than the same well [4]. LINCS L1000 is the deepest public chemistry catalog by compound count, with bulk transcriptional profiles for tens of thousands of compounds in a 1,000-gene reduced expression space [2]. Cell Painting is the standard for ML-on-imaging, with the JUMP consortium and Recursion-derived public sets providing single-cell or whole-well morphological profiles at high library coverage [5].

The current Z-Screen canonical dataset contributes 615,793 repaired RNA profiles spanning 12 combinatorial libraries and 4 cell lines, with chemistry-by-design provenance, plus matched per-well imaging on substantial subsets (companion paper1). Direct comparison places Z-Screen on a different axis from any of those references. Tahoe-100M is currently larger by a wide margin on raw RNA scale. scGeneScope remains the strongest public multimodal benchmark on raw treatment-matched accuracy. L1000 leads on raw small-molecule compound count. Cell Painting leads on imaging throughput. Z-Screen, even as a small early pilot, is the most chemistry-design-relevant atlas in the current public set: every well carries full building-block and SMILES provenance, the readout produces a learnable chemistry-to-transcriptome map at present scale, and the platform is engineered to scale by orders of magnitude. The contribution is design relevance and growth headroom, not raw atlas size.

The positioning has economic implications. A discovery team using L1000 can search compound space against a transcriptomic reference but cannot decompose a hit into the structural pieces that drove the response. A team using Cell Painting can scale to tens of millions of wells but cannot recover mechanism from morphology alone. Z-Screen is the only platform we know of that runs combinatorial chemistry, low-pass mRNA sequencing, and matched imaging in the same micro-well (each well containing 5 to 10 cells, profiled as a pseudobulk rather than at single-cell resolution), with chemistry decomposed into building-block coordinates. That design point converts a screen from a search tool into a model of structure-activity at the transcriptional level.

To anchor that positioning quantitatively, compound and cell-line overlap was computed between the Z-Screen canonical control panel and the two most directly comparable atlases, and that overlap was used to run an explicit cross-platform signature concordance test against LINCS L1000 (**Benchmarking/** in this repository). Of 47 named Z-Screen control compounds, 14 (29.8%) appear in the LINCS L1000 CMap LINCS 2020 build (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, AZD-3463, BMS-509744, BGT226, ZM-336372). Three of four Z-Screen cell lines (HEK293, A549, THP1) are present in LINCS L1000 with sample coverage; only H1650 is absent. Tahoe-100M’s drug metadata covers 4 of the 47 named compounds, with A549 as the only Z-Screen cell line in Tahoe-100M’s panel.

For the L1000 concordance test, Z-Screen rank signatures were restricted to the 12,328-gene LINCS-visible universe so the two platforms are scored in the same gene space, then Spearman rank correlation was computed between Z-Screen up/down rank signatures and LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742) for each (compound, cell line) pair where both platforms have adequate coverage. Across 11 such pairs (10 in A549, 1 in HEK293), all 11 produced positive cross-platform Spearman correlation. Median rho was 0.490, mean 0.393, and 8 of 11 pairs cleared rho = 0.2. Five compounds reached empirical p < 0.048 against a same-cell-line unmatched-compound permutation null: ZM-336372 (rho = 0.625, p = 0.022), rucaparib (rho = 0.530, p = 0.023), sorafenib in A549 (rho = 0.505, p = 0.023), veliparib (rho = 0.490, p = 0.046), and crizotinib (rho = 0.679, p = 0.048). The cell-line-pooled comparison in HEK293, where sorafenib in HEK293T reached rho = 0.597, separated matched from unmatched compound pairings at one-sided Mann-Whitney p = 0.009. Two compounds (palbociclib at rho = 0.182, GSK126

at $\rho = 0.046$) fall in the bulk of the unmatched distribution; the GSK126 result is consistent with the documented slow transcriptional onset of EZH2 inhibition under LINCS’s 6 h treatment window. Z-Screen rank signatures therefore recover compound-specific transcriptional structure that an independent reference platform reports, in a gene-restricted concordance test that uses no platform-specific calibration.

Discussion

Z-Screen behaves as a learnable chemistry-to-transcriptome platform on data from a small number of pilot experiments. Control compounds are stable, held-out transcriptomic states can be predicted from chemistry, reference-like biological programs recur in the libraries (with a strong background-calibrated mimicry hit in HEK293), building-block programs partially transfer across cell lines, and the strongest systems show meaningful gains from combining building-block provenance with full structural descriptors.

There is a clear hierarchy in the current evidence. RNA carries the strongest predictive signal today. Imaging is directionally positive in this analysis and considerably stronger when calibrated for same-well matched readout (companion paper1). The imaging branch is best framed as an additive layer on a working chemistry-to-RNA model at this stage, with room to grow as the imaging workflow improves.

The most informative next experiments follow directly. The first is greater depth in sparse libraries, especially systems that underperform when coverage per building block is low. The second is a prospective cross-library benchmark that asks whether a model trained on one scaffold family can predict transcriptomic programs in another; companion paper3 sets this up explicitly. The third is continued improvement in image consistency so that morphology can contribute more strongly to the readout. Each of these compounds a signal already present, on a screening platform engineered to scale by orders of magnitude beyond the current dataset.

For a discovery team, combinatorial chemistry inside a 50,000-well one-bead-one-compound microchip with low-pass mRNA sequencing produces a structured chemistry-to-transcriptome map at compound resolution rather than a sparse hit catalog. The economic implication is that a screen run on this platform leaves behind a reusable model rather than only a list of wells.

Methods

Dataset and preprocessing

The analysis uses the repaired canonical Z-Screen workspace, which contains RNA profiles, scVI latent coordinates, chemistry annotations, and selected image-derived features across 12 combinatorial libraries and 4 cell lines. Named controls were retained for reproducibility and mimicry analyses. Transcriptomic models were evaluated on library and cell-line subsets with sufficient compound coverage. The 72 ZIC004 / A549 rows without valid scVI latent coordinates were retained in the RNA count table but excluded from latent-space modeling; this is the source of the small mismatch

between `canonical_obs.parquet` (615,793 rows) and `canonical_scvi_latents.parquet` (615,721 rows with valid coordinates) noted in the package README.

scVI latent space rather than gene-level expression

Most analyses operate on a 32-dimensional scVI latent space rather than full gene-level expression vectors, for two reasons. First, low-pass per-well mRNA-seq (each well is a pseudobulk of 5 to 10 cells) is dominated by sampling noise in any single gene; scVI’s variational latent absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments in companion paper⁴ require. Where gene-level signal is the appropriate unit (control mimicry, gene-program inspection in companion paper¹) centered pseudobulk log-CPM is used directly.

Control reproducibility

Control split-half reproducibility was computed from repeated random partitions of wells belonging to the same reference compound. Similarity was summarized with cosine similarity between pseudo-replicate centroids in scVI latent space. Repeated partitioning rather than a single split was used to keep the variance of the metric low.

Building-block predictive modeling

Held-out tuple analyses fit additive and additive-plus-pair building-block models to transcriptomic centroids and evaluated them on compounds excluded from training by tuple identity. The choice of additive plus pair (rather than a high-order interaction model) reflects the present library coverage: pair-level interactions are well sampled in ZEL024, while higher-order terms run into combinatorial explosion at present scale. Performance was summarized with median cosine similarity to the measured transcriptomic target and mean squared error reduction relative to a centroid baseline. Companion paper³ extends this analysis to the harder building-block holdout and cross-library transfer regimes.

Chemistry-to-RNA prediction

Chemistry-only benchmarks compared models using a 256-dim per-compound chemistry embedding, building-block one-hot identities, or their concatenation. The chemistry embedding is a private fixed-seed Gaussian random projection from $\mathbb{R}^{2048}$ to $\mathbb{R}^{256}$ applied to per-compound 2048-bit ECFP4 (Morgan radius 2) fingerprints, then L2-normalized; this representation preserves Tanimoto-like geometry under cosine similarity (Johnson-Lindenstrauss) while being mathematically irreversible to substructure information without the projection matrix, which is held privately in the NDA bundle. A fixed deterministic projection was used in preference to a learned chemistry encoder because the projection is reproducible across runs, requires no separate training step, and decouples the published chemistry features from the underlying SMILES strings. Outputs were scored against measured transcriptomic centroids using cosine similarity and mean squared error reduction relative to a centroid baseline.

Control mimicry, cross-control normalization, and cross-cell building-block analysis

Control mimicry identified library tuples whose transcriptomic centroids were most similar to named reference compounds within the same library and cell line. Mimicry hits were calibrated against the per-cell-line cross-control similarity background (median, 95th percentile, 99th percentile cosines among unrelated reference compounds in the same cell line), so that “high” similarity is interpreted relative to the cell-line baseline rather than against zero. Cross-cell building-block consistency compared the direction of per-building-block transcriptomic programs across paired cell lines within the same library.

Additive multimodal benchmark

The multimodal probe followed a conservative additive framing: image features, RNA features, and their concatenation were evaluated on held-out class prediction tasks constructed from the available image-linked data. Performance was summarized with balanced accuracy and mean average precision. The additive frame was chosen because it produces a result traceable to a specific feature axis; cross-attention or contrastive multimodal models can in principle do better, but the gain is harder to attribute to one modality.

External benchmark context

Tahoe-100M, scGeneScope, L1000, and Z-Screen statistics were assembled into a single state-of-the-art comparison table covering RNA scale, image scale, cell-line coverage, perturbation diversity, pairing structure, and a representative result per dataset. The comparison was used only for positioning, not to claim parity on raw multimodal balanced accuracy.

Limitations

The current analysis is based on a small number of pilot experiments rather than a fully saturated atlas. RNA carries the strongest signal today; the image branch is directionally positive and depends on derived feature artifacts rather than raw microscopy in this shareable package. Control mimicry is calibrated against same-cell-line cross-control backgrounds, but the strongest discovery-style hits still require orthogonal experimental validation before they should be treated as confirmed mechanisms of action. Cross-library and prospective scaffold-family transfer are set up by the present results but are not claimed as solved here.

Data availability

Data tables, derived image features, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper2/`, `data/ZScreen_Canonical_Dataset/`, and `data/paper2_artifacts/`. Raw microscopy images are not included in the shareable bundle; the paper2 image-linked analyses use the derived parquet feature files documented in the package README. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper2/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

References

1. Lopez R, Regier J, Cole MB, et al. Deep generative modeling for single-cell transcriptomics. *Nat Methods*. 2018;15:1053-1058. doi:10.1038/s41592-018-0229-2
2. Subramanian A, Narayan R, Corsello SM, et al. A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles. *Cell*. 2017;171(6):1437-1452.e17. doi:10.1016/j.cell.2017.10.049
3. Tahoe Therapeutics. Tahoe-100M: A 100-million single-cell perturbational atlas across 50 cancer cell lines. Dataset card and February 2025 release. <https://huggingface.co/datasets/tahoe-bio/Tahoe-100M>
4. Dapello J, Nassar M, Eksi R, et al. scGeneScope: A treatment-matched single-cell imaging and transcriptomics dataset and benchmark for treatment response modeling. *NeurIPS 2025 Datasets and Benchmarks*. OpenReview. <https://openreview.net/pdf/f7d541dae38bcf88a79789a4c6440aadfec123c7>
5. Bray MA, Singh S, Han H, et al. Cell Painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes. *Nat Protoc*. 2016;11:1757-1774. doi:10.1038/nprot.2016.105
6. Dixit A, Parnas O, Li B, et al. Perturb-Seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell*. 2016;167(7):1853-1866.e17. doi:10.1016/j.cell.2016.11.038